



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)  
REVIEW TEST  
10-07-2021

## JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

### PAPER-1

Time : 2:45 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

#### GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

#### OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

#### DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take  **$g = 10 \text{ m/s}^2$**  unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

**SOME USEFUL CONSTANTS**

**Atomic No.** : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

**Atomic masses** : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- |                                    |                                                              |
|------------------------------------|--------------------------------------------------------------|
| • Boltzmann constant               | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$                   |
| • Coulomb's law constant           | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$                   |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$  |
| • Speed of light in vacuum         | $c = 3 \times 10^8 \text{ ms}^{-1}$                          |
| • Stefan-Boltzmann constant        | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$                        |
| • Permeability of vacuum           | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$                |
| • Permittivity of vacuum           | $\epsilon_0 = \frac{1}{\mu_0 c^2}$                           |
| • Planck constant                  | $h = 6.63 \times 10^{-34} \text{ J-s}$                       |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

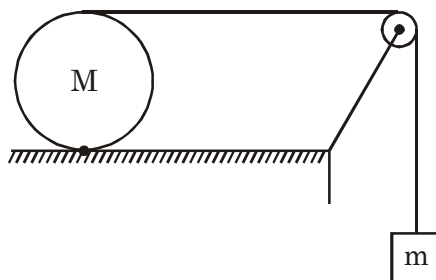
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :  
*Full Marks* : +3 If only the bubble corresponding to the correct option is darkened.  
*Zero Marks* : 0 If none of the bubbles is darkened.  
*Negative Marks* : -1 In all other cases

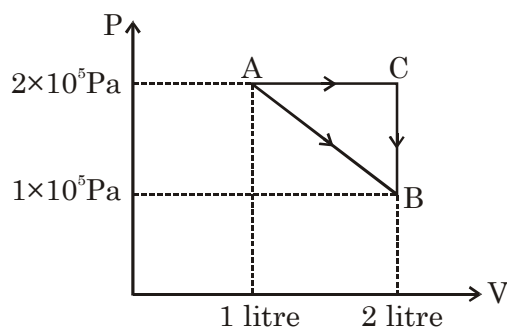
1. A massless inextensible string is wrapped over a cylinder of radius 20 cm and mass of 2kg. It is connected to a hanging block of mass 1 kg. If the surface is smooth, the cylinder reaches a distance  $\ell$  in time  $t_1$ . If the surface is sufficiently rough, the cylinder reaches the distance  $\ell$  in time  $t_2$ . Then  $\frac{t_1}{t_2} =$



- (A) 1                      (B)  $\sqrt{\frac{7}{10}}$                       (C)  $\sqrt{\frac{10}{7}}$                       (D)  $\frac{1}{\sqrt{2}}$

Space for Rough Work

2. An equi-convex lens ( $\mu = 1.5$ ) is combined with an equi concave lens ( $\mu = 1.3$ ). The radii of curvature of all surfaces is the same. (Assume thin lenses)
- (A) The combination behaves like a converging lens in medium of refractive index 1.4 and diverging lens in a medium of refractive index 1.6.
- (B) The combination behaves like a converging lens in a medium of refractive index 1.2 as well as 1.4.
- (C) The combination behaves like a diverging lens in medium of refractive index 1.2 as well as 1.6.
- (D) The combination behaves like a diverging lens in medium of refractive index 1.4 and converging lens in a medium of refractive index 1.6.
3. A certain substance (not an ideal gas) has a PV graph as a straight line AB when subjected to an adiabatic process. If the same substance goes through process  $A \rightarrow C \rightarrow B$ , what is the heat given to it ?



(A) 50 J

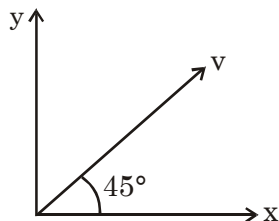
(B) 200 J

(C) 150 J

(D) 100 J

Space for Rough Work

4. A charge of  $+10\mu\text{C}$  and mass of  $1\text{gm}$  is moving with velocity of  $10^6\text{ m/s}$  at an angle of  $45^\circ$  to  $x$  &  $y$  axis as shown. It experiences a magnetic force along  $-ve$   $z$ -direction. When it is projected with  $10^6\text{ m/s}$  along  $+ve$   $z$ -direction, it experiences a magnetic force of  $10^{-2}\text{ N}$  in  $+ve$   $x$ -direction. The magnetic field  $\vec{B}$  is :



- (A)  $-10^{-3}\text{T}(\hat{i} + \hat{j})$       (B)  $10^{-3}(\hat{j} + \hat{k})$       (C)  $10^{-3}\hat{i}$       (D)  $-10^{-3}\text{T}\hat{j}$

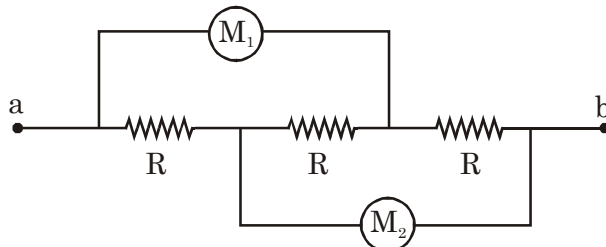
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Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.  
*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.  
*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.  
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.  
*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).  
*Negative Marks* : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. In the given circuit the potential difference between a & b is 300 V. The resistors are of  $100\ \Omega$  each.



- (A) If the upper meter  $M_1$  is ideal voltmeter & lower is ideal ammeter  $M_2$ , reading of voltmeter is 300 V.
- (B) If both meters are ideal voltmeters, they show 200 V each.
- (C) If both meters are ideal ammeters, they read 6A each.
- (D) If the upper meter  $M_1$  is ideal voltmeter and lower is ideal ammeter  $M_2$ , reading of ammeter is 3A.

Space for Rough Work

**ALLEN**

6. A plastic body of density  $\rho$  is to be held in air of density  $\rho_a$  first and then in water of density  $\rho_w$ . For holding, we need to apply a force on it.

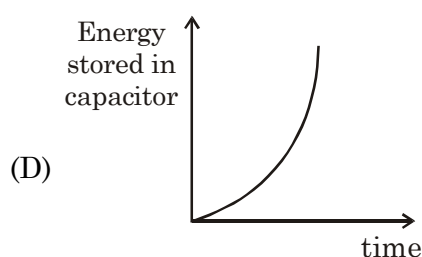
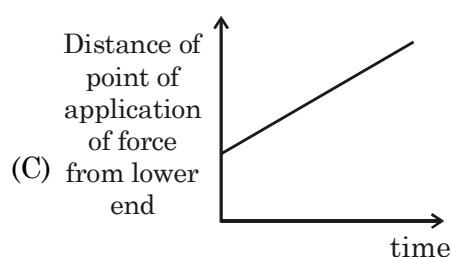
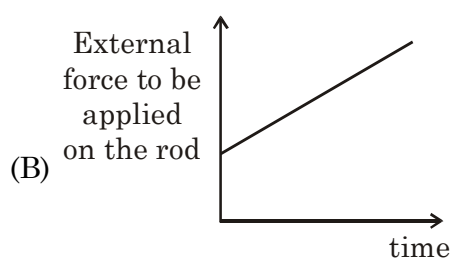
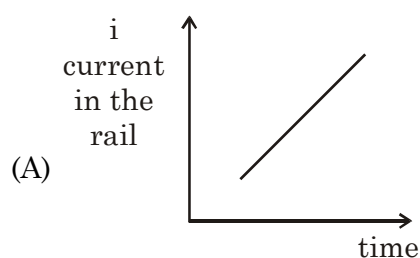
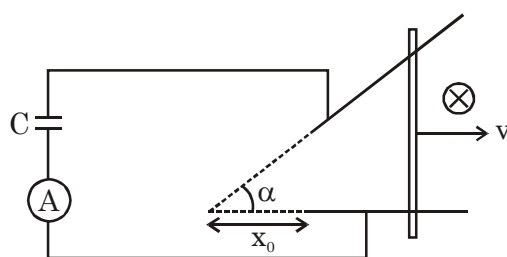
(A) It is easier to hold it in water if  $\rho > \rho_a$  &  $\rho < \frac{\rho_w + \rho_a}{2}$ .

(B) It is easier to hold it in water if  $\rho > \rho_w$ .

(C) It is easier to hold it in air if  $\rho < \rho_w$  &  $\rho < \frac{\rho_w + \rho_a}{2}$ .

(D) It is easier to hold it in water if  $\rho < \rho_w$  &  $\rho > \frac{\rho_w + \rho_a}{2}$ .

7. A long metallic rod of mass  $m$  is being moved on two smooth horizontal metallic rails with constant velocity  $v$ . The angle between the rails is  $\alpha$ . A uniform magnetic field ( $B$ ) in vertical direction is present as shown. The capacitor is initially uncharged. The resistance of circuit is zero. Choose **CORRECT** graph(s) :-



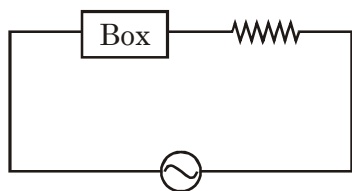
Space for Rough Work

8. A radioactive nucleus A decays to C after emitting two  $\alpha$  and three  $\beta^-$  particles. Another nucleus B decays to C by emitting one  $\alpha$  and five  $\beta^-$  particles. Half lives of A & B are 1 hr and 2hr respectively and intermediaries have negligible half lives. At time  $t = 0$ ; population of A is  $4N_0$ , of B is  $N_0$  and of C is zero.
- (A) Atomic mass number of A – Atomic mass number of B = 4  
 (B) Atomic number of A – Atomic number of B = – 2
- (C) When A & B have equal population, population of C =  $\frac{9N_0}{2}$ .
- (D) The decay rate of A & B are equal at  $t = 6\text{hr}$ .
9. When a narrow resonance column apparatus is filled with ozone gas ( $\gamma = 4/3$ ), it resonates in 1<sup>st</sup> resonance with a tuning fork. When it is filled with nitrogen gas ( $N_2$ ), it resonates with the same tuning fork in 1<sup>st</sup> resonance when water column is shifted down by 10 cm.
- (Take  $\sqrt{5} = 2.25$ )
- (A) Wavelength of sound in Ozone is approximately 1m.  
 (B) Wavelength of sound in Nitrogen is approximately 1.6 m.  
 (C) For second resonance in Ozone gas, the water column should be shifted down by 60 cm.  
 (D) For second resonance in Nitrogen gas, the water column should be shifted down by 80 cm.

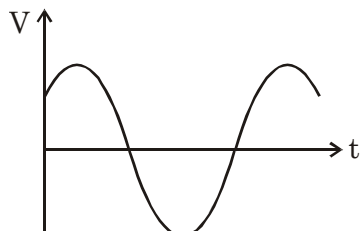
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**Space for Rough Work**

10. In an AC circuit, box may contain an inductor or a capacitor.

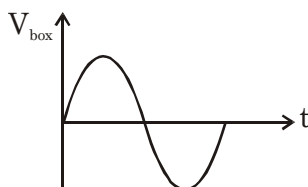


The voltage across source is shown by given graph

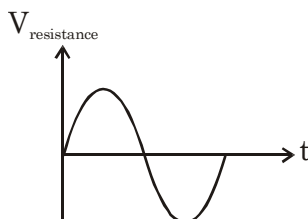


Here  $V_{\text{box}}$  versus time &  $V_{\text{resistance}}$  versus time graphs are plotted. Choose the possible **CORRECT** alternative(s) :

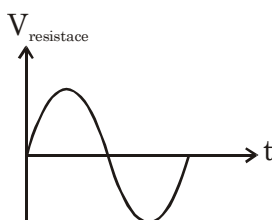
(A) If box contains inductor,



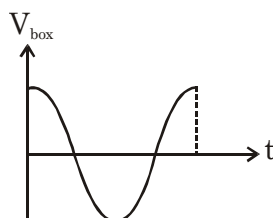
(B) If box contains inductor,



(C) If box contains capacitor,



(D) If box contains capacitor,



Space for Rough Work

11. A body of mass  $m$  experiences a central force  $\vec{F} = -k\vec{r}$ . Here  $\vec{r}$  is the position from centre of force and  $k$  is a positive constant. It moves in a closed circular path such that its angular

momentum about centre of force is quantized  $L = \frac{nh}{2\pi}$ . Take potential energy = 0 at  $r = 0$ .

- (A) Its possible radius is quantized such that  $r \propto n^{1/2}$ .  
 (B) Its possible mechanical energy is quantized such that  $E \propto n$ .  
 (C) Its speed is quantized such that  $v \propto n^{1/2}$ .  
 (D) Its angular velocity is quantized such that  $\omega \propto n$ .

12. On a smooth horizontal surface a lightweight rigid rod of length  $L$  slides. At the ends of the rod are two point masses  $M$  and  $2M$ . At some time ( $t = 0$ ) the velocity of lighter mass is  $v\hat{i}$  and is directed in a direction perpendicular to the rod. Mass  $M$  at that instant is on positive  $y$  axis and  $2M$  on negative  $y$  axis. The velocity of the heavier mass is  $2v\hat{i}$ . Choose the correct statement(s) :

- (A) The velocity of the lighter mass as a function of time is given by

$$\frac{1}{3}v\hat{i} + \frac{2}{3}v\left(\cos\left(\frac{vt}{L}\right)\hat{i} + \sin\left(\frac{vt}{L}\right)\hat{j}\right)$$

- (B) The velocity of the heavier mass as a function of time is given by

$$\frac{5}{3}v\hat{i} + \frac{1}{3}v\left(\cos\left(\frac{vt}{L}\right)\hat{i} + \sin\left(\frac{vt}{L}\right)\hat{j}\right)$$

- (C) The velocity of the centre of mass is given by  $\frac{5}{3}v\hat{i}$

- (D) The magnitude of tension in the rod is  $\frac{2}{3L}Mv^2$

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Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

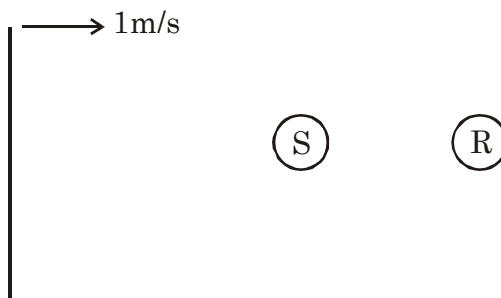
**For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

| + |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| ● | ● | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | ● | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |

| - |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| ● | ● | ● | 0 | 0 | ● | ● | ● | ● | ● |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | ● | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |

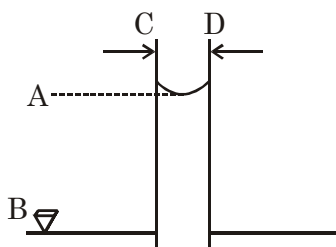
- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +3 If **ONLY** the correct numerical value is entered as answer.  
*Zero Marks* : 0 In all other cases.

- Electromagnetic waves of wavelength  $\lambda = 500$  nm are incident on a YDSE apparatus. Both slits are of same width. Here  $d = 3$  mm,  $D = 2$  m. Intensity at central maxima is  $1 \text{ W/m}^2$ , what is the intensity (in  $\text{W/m}^2$ ) at a point 1 mm above maxima of order 1?
- A point source and a receiver are stationary. Source sends a wave of frequency 256 Hz, which directly reaches to receiver and also after being reflected from a wall. The wall is moving at speed of 1 m/s. Find beat frequency (in Hz) heard by receiver. Velocity of sound = 321 m/s.

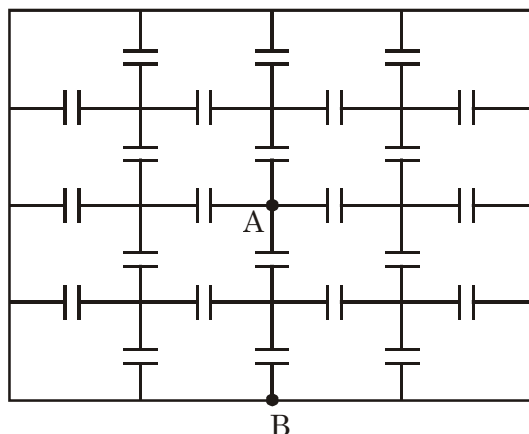


Space for Rough Work

3. Surface tension of a liquid is being estimated by measuring height of liquid rises in a capillary. The travelling microscope has vernier scale whose reading for water level B is 2.3 cm on main scale and 4<sup>th</sup> vernier scale division coinciding. The reading for lower level of meniscus A is 4.7 cm on main scale and 9<sup>th</sup> vernier scale division coinciding. For measuring the diameter, the left end of the capillary C measures 3.1 cm on main scale and 1<sup>st</sup> vernier scale division coinciding. The right end of capillary D measures 3.2 cm on main scale and 4<sup>th</sup> vernier scale division coinciding. Here angle of contact = 0°, density is  $10^3 \text{ kg/m}^3$  and acceleration due to gravity =  $10 \text{ m/s}^2$  and 10 vernier scale divisions = 9 main scale divisions. What is value of surface tension (in N/m)?

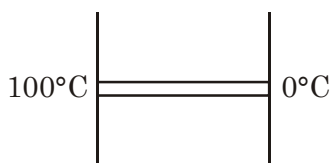


4. A charge of  $8\mu\text{C}$  is given to point A and point B is earthed. If each capacitance is  $1 \mu\text{F}$ , what is the total energy (in  $\mu\text{J}$ ) of the system ?



Space for Rough Work

5. The charge density inside a nucleus is given by  $\rho = \rho_0 \left( 1 - \frac{r^2}{a^2} \right)$  where  $r \leq a$ ,  $a$  = radius of nucleus = 3 fm and  $\rho_0 = 5 \times 10^{25} \text{ C/m}^3$ . The electric field is maximum at  $r = r_0$ . Find  $r_0$  (in fm) and fill it in OMR sheet.
6. Two rigid walls at a distance of 1m act as a heat reservoir at  $100^\circ\text{C}$  and  $0^\circ\text{C}$ . A well lagged uniform rod having length of 1m at  $0^\circ\text{C}$  is held between both the walls. It is found that when the system reaches steady state, the supporting force can be removed and limiting friction acts on the rod at both the ends. Here  $\alpha$  for rod =  $10^{-5}/^\circ\text{C}$ ,  $Y = 2 \times 10^{11} \text{ N/m}^2$ ,  $\mu_{\text{walls}} = 0.3$  and area =  $4\text{mm}^2$ . What is the mass of the rod (in Kg) ?



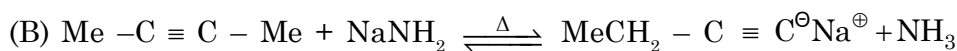
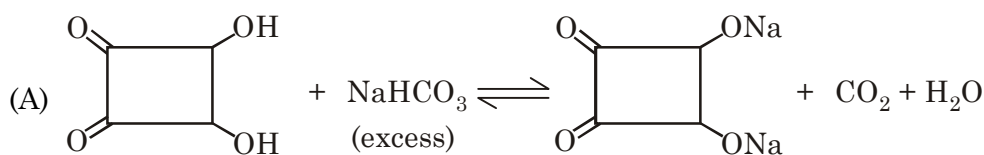
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 12)

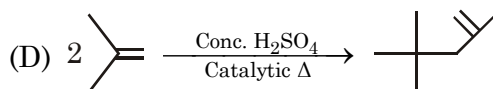
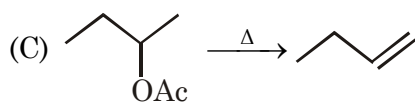
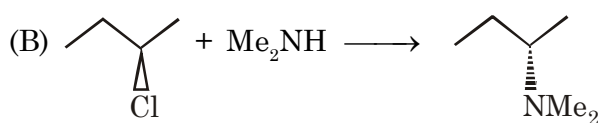
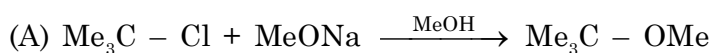
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1. Select reaction(s) which will favour in forward direction

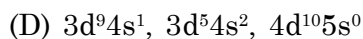
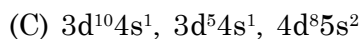


(D) None of these

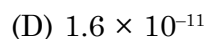
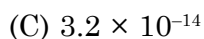
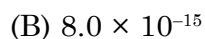
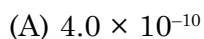
2. Select an option having incorrect major product?



3. Choose **CORRECT** outer electronic configuration of Cu, Cr & Pd are respectively.



4. The solubility of  $\text{Ag}_2\text{CO}_3$  in water is  $5520 \mu\text{g/L}$ .  $K_{sp}$  of  $\text{Ag}_2\text{CO}_3$  is ( $\text{Ag} = 108$ )



Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
  - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
  - For each question, choose the correct option(s) to answer the question.
  - Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.  
*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.  
*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.  
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.  
*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).  
*Negative Marks* : -1 In all other cases.
  - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.
- 
5. Which of the following atom/s has low ionization energy than Nitrogen atom.  
 (A) Li (B) F (C) O (D) Ne
6. Choose **CORRECT** statements.  
 (A) The bond dissociation energy order  $C_2 > C_2^+ > C_2^-$   
 (B) Formation of  $O_2^+$  from  $O_2$  electron is removed from gerade bonding molecular orbital  
 (C) HOMO of  $F_2$  is gerade while LUMO of  $Cl_2$  is ungerade  
 (D) In Halogen colour intensity increases  $F_2$  to  $I_2$
7. Which of the following statement(s) is/are correct about coagulation ?  
 (A) Tanning of leather is an example of mutual coagulation.  
 (B) Delta formation is an example of coagulation by electrolyte.  
 (C) Cottrell precipitator coagulate by electrophoresis.  
 (D) Lyophilic colloids may be coagulated by adding other suitable solvent.

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Space for Rough Work

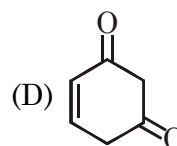
8. The molar conductance of  $\text{H}^+$  ion and  $\text{CH}_3\text{COO}^-$  ion at 298K and at infinite dilution are  $3.474 \times 10^{-2} \text{ ohm}^{-1} \text{ m}^2 \text{ mol}^{-1}$  and  $1.351 \times 10^{-2} \text{ ohm}^{-1} \text{ m}^2 \text{ mol}^{-1}$ , respectively. The resistance of 0.1 M  $\text{CH}_3\text{COOH}$  solution is 2000 ohm at 298K in a conductivity cell of cell constant  $20 \text{ m}^{-1}$ . The correct information(s) related with the given parameters at 298K is/are- ( $1\text{F} = 96500 \text{ C/mol}$ )
- (A)  $\Lambda_m^0(\text{CH}_3\text{COOH}) = 4.825 \times 10^{-2} \text{ ohm}^{-1} \text{ m}^2 \text{ mol}^{-1}$
- (B)  $\kappa = 1.0 \times 10^{-2} \text{ ohm}^{-1} \text{ m}^{-1}$
- (C)  $\Lambda_m = 1.0 \times 10^{-4} \text{ ohm}^{-1} \text{ m}^2 \text{ mol}^{-1}$
- (D)  $[\text{H}^+] = 2.07 \times 10^{-3} \text{ M}$

9. Which of the following will give alkane as product with  $\text{RMgCl}$ ?

(A)  $\text{PhOMe}$

(B)  $\text{Me}_3\text{N}$

(C)  $\text{Me}_3\text{C} - \text{Cl}$



10. Neopentane can be prepared in good yield by

(A) Catalytic reduction

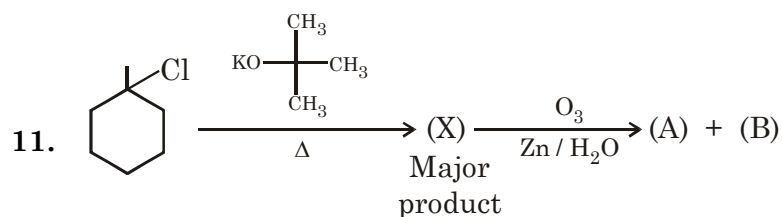
(B) Wolf-Kischner's reduction

(C) Wurtz reaction

(D) Decarboxylation using soda lime

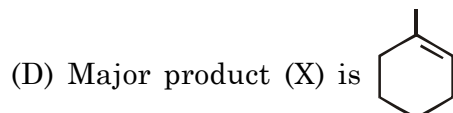
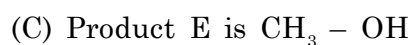
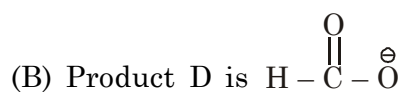
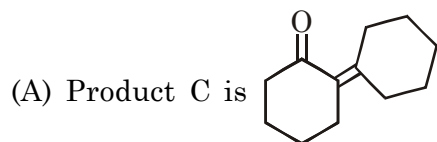
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Space for Rough Work



Product A and B are separated and then heated with NaOH. A gave C whereas B gave D and E

Correct statement(s) is/are :



12. Choose **CORRECT** statements.

(A) The dark blue colour of the solution formed when excess of ammonia is added to a solution of copper(II) sulphate is due to the formation of diamagnetic  $[\text{Cu}(\text{NH}_3)_4]^{2\oplus}$  ion

(B)  $[\text{Cr}(\text{H}_2\text{O})_6]^{3\oplus}$  have more paramagnetism than  $[\text{Ni}(\text{H}_2\text{O})_6]^{2\oplus}$

(C)  $[\text{Ni}(\text{CO})_4]$  is tetrahedral and paramagnetic complex

(D)  $[\text{Ni}(\text{DMG})_2]$  is low spin square planar complex

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

**For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

| + |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| ● | ● | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ |
| ① | ① | ① | ① | ① | ① | ① | ① | ① | ① |
| ② | ② | ② | ② | ● | ② | ② | ② | ② | ② |
| ③ | ③ | ③ | ③ | ③ | ③ | ③ | ③ | ③ | ③ |
| ④ | ④ | ④ | ④ | ④ | ④ | ④ | ④ | ④ | ④ |
| ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ |
| ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ |
| ⑦ | ⑦ | ● | ● | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ |
| ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ |
| ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ |

| - |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| ● | ● | ● | ○ | ○ | ○ | ○ | ○ | ○ | ○ |
| ① | ① | ① | ① | ① | ① | ① | ① | ① | ① |
| ② | ② | ② | ② | ● | ② | ② | ② | ② | ② |
| ③ | ③ | ③ | ③ | ③ | ③ | ③ | ③ | ③ | ③ |
| ④ | ④ | ④ | ④ | ④ | ④ | ④ | ④ | ④ | ④ |
| ⑤ | ⑤ | ⑤ | ⑤ | ● | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ |
| ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ |
| ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ |
| ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ |
| ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ |

- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +3 If ONLY the correct numerical value is entered as answer.  
*Zero Marks* : 0 In all other cases.

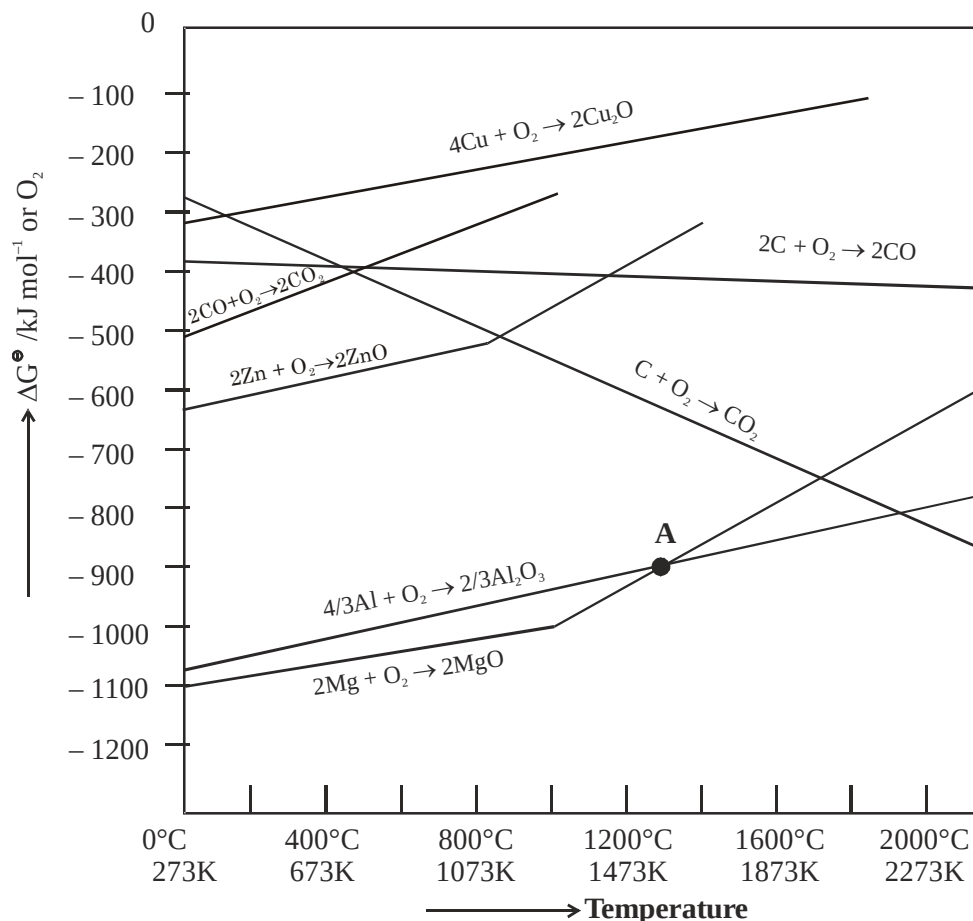
- The normal boiling point of pure water is 373.15 K. Calculate the boiling point (in K) of an aqueous solution containing 11.7 gm NaCl and 34.2 gm sugar,  $C_{12}H_{22}O_{11}$ , in 260 gm water. For water,  $K_b = 0.52$  K-Kg/mol. [Atomic mass of Na = 23, Cl = 35.5]
- The reaction :  $A(g) \rightarrow 2B(g) + C(l)$  follows first order kinetics. The reaction is started with A(g) only. Calculate the rate constant of reaction (in  $\text{min}^{-1}$ ) from the following data :  
( $\ln 2 = 0.6931$ )

| Time (min.)                    | 0     | 10    | $\infty$ |
|--------------------------------|-------|-------|----------|
| Total pressure of system (bar) | 0.080 | 0.125 | 0.165    |

- One mole of non-ideal gas undergoes a process (10 L, 2 atm, 300 K) to (10L , 5 atm, 800K). The molar enthalpy change of gas in this process (in L-atm/mole) is  
(Given :  $C_{v,m} = 3.2R$ ,  $C_{p,m} = 4.1 R$  ,  $R = 0.08$  L-atm/K-mole).

Space for Rough Work

4. How many of the following are more reactive than benzene for electrophile substitution reaction?
- (1) Anisole (2) Acetanilide (3) Carboic acid (4) TNT  
 (5) Mesitylene (6) Benzaldehyde (7) Acetophenone (8) Phenyl benzoate  
 (9) Meta-Xylene (10) Ethyl benzoate
5. Total number of plot(s) in which slope is not correctly represented.



6. Total number of anion which give white ppt. with aqueous solution of  $\text{AgNO}_3$ .  
 $\text{S}^{2-}$ ,  $\text{HSO}_3^\ominus$ ,  $\text{SO}_3^{2-}$ ,  $\text{S}_2\text{O}_3^{2-}$ ,  $\text{CH}_3\text{COO}^\ominus$ ,  $\text{Cl}^\ominus$ ,  $\text{I}^\ominus$ ,  $\text{BO}_3^{3-}$

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :  
*Full Marks* : +3 If only the bubble corresponding to the correct option is darkened.  
*Zero Marks* : 0 If none of the bubbles is darkened.  
*Negative Marks* : -1 In all other cases

1. The sum of the series  $\sin x \sec 3x + \sin 3x \sec 3^2 x + \sin 3^2 x \sec 3^3 x + \dots$  upto  $n$  terms is  
 (A)  $\frac{1}{2} [\cot 3^n x - \cot x]$  (B)  $\frac{1}{2} [\tan 3^n x - \tan x]$   
 (C)  $[\tan 3^n x - \tan x]$  (D)  $\cot 3^n x - \cot x$
2. If  $ax + y - 16 = 0$  and  $x + ay - 6 = 0$  represent two adjacent sides of a rhombus whose diagonals intersect at  $(3, 5)$  then sum of possible value of  $a$  is  
 (A) 4 (B) 2 (C) -2 (D)  $-\frac{9}{4}$
3. Locus of a variable point  $P(x, y)$  is given by  $x^3 + y^3 + 3xy = 1$  where  $(x, y) \neq (-1, -1)$ . The equation of circle touching the locus of  $P$  at  $(-1, 2)$  and passing through  $(1, -2)$  is given by  
 (A)  $x^2 + y^2 + 2x + y - 5 = 0$  (B)  $x^2 + y^2 + 12x + 6y - 5 = 0$   
 (C)  $x^2 + y^2 + 4x + 2y - 5 = 0$  (D)  $x^2 + y^2 - 4x - 2y - 5 = 0$
4. If  $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \frac{\sin([x]\pi)}{x^2 + 2x + 3} + 2x - 1 + \sqrt{x(x-1)} + \frac{1}{4}$  (where  $[x]$  denotes greatest integral value less than or equal to  $x$ ) denotes a function, then number of real solutions of equation  $f(x) = f^{-1}(x)$  is  
 (A) 0 (B) 1 (C) 2 (D) 3

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
  - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
  - For each question, choose the correct option(s) to answer the question.
  - Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.  
*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.  
*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.  
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*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).  
*Negative Marks* : -1 In all other cases.
  - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.
- 
5. Let  $f$  be a twice differentiable function defined in  $[-4,4]$  such that  $f'(-4) = 10$ ,  $f'(4) = -14$  and  $f''(x) \geq -3 \forall x \in [-4,4]$ . If  $g(x) = \int_0^x f(t) dt$  and  $f(0) = 0$ , then
- (A)  $g(x)$  increases in  $(-4,0)$   
 (B)  $g(x)$  decreases in  $(0,4)$   
 (C) graph of  $g(x)$  is concave down in  $(0,4)$   
 (D)  $f(x)$  has maximum value at  $x = 2$

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Space for Rough Work

6. Which of the following can be antiderivative of  $f(x) = \frac{x^4 + 1}{x^6 + 1}$  ? (Where 'C' is the constant of integration)

(A)  $\tan^{-1} x + \frac{1}{3} \tan^{-1}(x^3) + C$

(B)  $\tan^{-1}\left(x - \frac{1}{x}\right) - \frac{2}{3} \tan^{-1}(x^3) + C$

(C)  $\frac{1}{3} \tan^{-1} x + \tan^{-1}(x^3) + C$

(D)  $\frac{2}{3} \tan^{-1}\left(x - \frac{1}{x}\right) - \tan^{-1}(x^3) + C$

7. Let  $z - c = \rho \left( \frac{1 + it}{1 - it} \right)$  where  $c$  is a complex number,  $\rho$  and  $t$  are real parameters then

(A) if  $z = c$  and  $\rho = 0$  then  $t$  is undefined

(B) if  $z - c \in \mathbb{R}$  then  $\rho = z - c$  and  $t = 0$

(C) if  $z \neq c$  then  $\rho = \pm |z - c|$  and  $t = \frac{\text{Im}(z - c)}{\pm |z - c| + \text{Re}(z - c)}$

(D) If  $z \neq c$  then  $\rho = \pm |z - c|$  and  $t = \frac{\text{Im}(z - c)}{\pm |z - c| - \text{Re}(z - c)}$

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**Space for Rough Work**

8. Let  $|r| < 1$  and  $S_n = \sum_{k=1}^{\infty} k^n r^k$ , then which of the following options is/are correct ?

(A)  $S_0 = \frac{1}{1-r}$       (B)  $rS_1 = (S_0)^2$       (C)  $S_{n+1} = r \frac{d}{dr}(S_n)$       (D)  $rS_2 = S_0^2 + 2S_0^3$

9. In a game show a contestant selects one of three doors. Behind one of the doors there is a prize and behind other two there is no prize. Each door has equal probability of having a prize. After contestant selects a door, the game show host who knows what's behind each of the doors, deliberately opens one of the remaining doors and reveals there is no prize behind it. The host then asks the contestant whether he want to SWITCH his choice to other unopened door or STICK to his original choice. Which of the following options is/are correct ?

(A) Probability of winning the prize is same whether contestant STICK or SWITCH

(B) Probability of winning the prize when contestant STICK, is  $\frac{1}{3}$

(C) Probability of winning the prize when contestant SWITCH, is  $\frac{2}{3}$

(D) It is advantageous for contestant to SWITCH than to STICK

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**Space for Rough Work**

10. Consider a system of linear equations in three variables

$$-x + y + z = a$$

$$x - y + z = b$$

$$x + y - z = c$$

then

$$(A) \ x + y = \frac{a+b}{2} + c \quad (B) \ y + z = a + \frac{b+c}{2} \quad (C) \ x + z = \frac{a+c}{2} + b \quad (D) \ x + y + z = a + b + c$$

11. Two tangents are drawn from point  $P(\alpha, \beta)$  to the hyperbola.  $3x^2 - 2y^2 = 6$  and are inclined at angles A and B to positive direction of x-axis. Let  $\tan A \cdot \tan B = k$ , then

$$(A) \text{ if } k = 2 \text{ then } \beta^2 = 2\alpha^2 + 7$$

$$(B) \text{ if } k = 2 \text{ then } \beta^2 = 2\alpha^2 - 7$$

$$(C) \text{ if } k = 3 \text{ then } \beta^2 = 3\alpha^2 - 9$$

$$(D) \text{ if } k = 3 \text{ then } \beta^2 = 3\alpha^2 + 9$$

12. If  $2^{n-1}T_n = {}^{2n}C_0 + 3({}^{2n}C_2) + 3^2({}^{2n}C_4) + \dots + 3^n({}^{2n}C_{2n})$ ,  $n \in \mathbb{N}$ , then

$$(A) \ T_n, T_{n+1}, T_{n+2} \text{ are in AP}$$

$$(B) \ T_n, 2T_{n+1}, T_{n+2} \text{ are in AP}$$

$$(C) \ T_4 = 192$$

$$(D) \ T_4 = 194$$

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**Space for Rough Work**

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

**For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

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- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +3 If **ONLY** the correct numerical value is entered as answer.  
*Zero Marks* : 0 In all other cases.

1. Evaluate  $\lim_{x \rightarrow 0} \left( \frac{\tan x}{x} \right)^{\left( \frac{\tan x}{x - \tan x} \right)}$  (consider  $e = 2.7$ )
2. If  $f(x) = \frac{1}{2^{n-1}}$  for  $\frac{1}{2^n} < x \leq \frac{1}{2^{n-1}}$ ,  $x \in [0, 1]$ ,  $n \in \mathbb{N}$  and  $f(0) = 0$ . Find the value of  $\int_0^1 f(x) dx$
3. If a curve represented by differential equation  $(\tan^{-1}y - x)dy = (1 + y^2)dx$ , passes through  $(0, \tan 1)$  and  $\left( \frac{\pi}{3} - 1, b \right)$ , then value of  $b$  is

Space for Rough Work

4. If  $\vec{a}$  and  $\vec{b}$  are perpendicular vectors with  $|\vec{a}| = 2, |\vec{b}| = 3$  and  $\vec{c} \times \vec{a} = \vec{b}$ , then the least value of  $|\vec{c} - \vec{a}|$
5. If  $\ell x + 13y + mz + n = 0$  is a plane through intersection of planes  $2x + 3y - z + 1 = 0$  and  $x + y - 2z + 3 = 0$  and is perpendicular to plane  $3x - y - 2z = 4$ , then  $\frac{\ell - n}{m}$  equals
6. Define permutation of set  $\{1, 2, 3, \dots, n\}$  to be sortable if upon cancelling one appropriate term of such permutation remaining  $n - 1$  terms are in increasing order. If  $f(n)$  is the number of such sortable permutations then  $\frac{f(5)}{f(4)}$  equals

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**QUESTION PAPER FORMAT AND MARKING SCHEME :**

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

| Section | Que. Type                                         | No. of Que. | Category-wise Marks for Each Question                                                   |                                                                                                          |                                         |                          | Maximum Marks of the section |
|---------|---------------------------------------------------|-------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------|------------------------------|
|         |                                                   |             | Full Marks                                                                              | Partial Marks                                                                                            | Zero Marks                              | Negative Marks           |                              |
| I(i)    | Single correct option                             | 4           | +3<br>If only the bubble corresponding to the correct option is darkened                | —                                                                                                        | 0<br>If none of the bubbles is darkened | –1<br>In all other cases | 12                           |
| I(ii)   | One or more correct option(s)                     | 8           | +4<br>If only the bubble(s) corresponding to all the correct option(s) is(are) darkened | +1<br>For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened | 0<br>If none of the bubbles is darkened | –1<br>In all other cases | 32                           |
| II      | Numerical Value Type (Up to second decimal place) | 6           | +3<br>If only the bubble corresponding to correct answer is darkened                    | —                                                                                                        | 0<br>In all other cases                 | —                        | 18                           |

NAME OF THE CANDIDATE .....

FORM NO. ....

I have read all the instructions and shall abide by them.

 \_\_\_\_\_  
Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

 \_\_\_\_\_  
Signature of the Invigilator